

MERGERS & ACQUISITIONS

EXAM STUDY GUIDE — HIGH DISTINCTION ANSWERS

BUSI70662 | Imperial College Business School | Summer 2026 | Part A: Concepts (60 marks)

CLASS 1 — Introduction to M&A

M&A Process: 5 Phases, 10 Steps

Q: Describe the overall M&A process (5 phases, 10 steps).

The M&A process runs across five phases and ten sequential steps, spanning from corporate strategy through to post-deal audit.

Phase	Steps	Timing
A — Corporate Strategy	1. Develop business plan 2. Develop M&A plan supporting business plan	Months / Years Before
B — Organising for M&A	3. Actively search for opportunities 4. Screen and prioritise targets	Months Before
C — Deal Making	5. First contact 6. Negotiation (refine valuation, deal structuring, DD, financing) 7. Develop integration plan 8. Closing — obtain approvals, execute closing	Weeks / Months
D — Integration	9. Implement post-closing integration plan	After Deal
E — Deal Audit	10. Evaluate deal post-closing and post-integration success	After Deal

Why Does M&A Matter?

Q: Why does M&A matter?

- There is a lot of it — M&A is inevitable in a firm's lifecycle. Any senior manager will face a transaction.
- It is a key piece of a capital allocation framework — alongside organic investment, buybacks, and dividends.
- It is complex, controversial, and attracts scrutiny — making analytical rigour essential.

Why Corporates Engage in M&A?

Q: Why do corporates engage in M&A?

- Synergies — most commonly cited empirical motivation. Cost savings, revenue benefits, or both.
- Strategic re-alignment — acquire new capabilities, markets, technologies, or distribution.
- Diversification — though empirically this has weak support and often results in conglomerate discount.
- Tax considerations — utilise tax shields, step-up asset bases.
- Misvaluation — acquiror believes target is undervalued ($Q\text{-ratio} < 1$ implies buying assets below replacement cost).
- Market power — consolidation to improve pricing power (though evidence shows efficiency gains more common).
- Management hubris / overconfidence — empirically significant cause of overpayment.
- Managerialism / agency problems — managers pursue growth for personal benefit rather than shareholder value.

Why M&A Comes in Waves?

Q: Why does M&A come in waves?

Merger waves occur when firms react to industry shocks — deregulation, new technologies, distribution disruptions, substitute products, or unpredictable events (e.g. a pandemic). Waves are amplified by: sustained high economic growth; rising stock markets (high equity currency); low or falling interest rates; high CEO confidence and low volatility. A secondary driver is contagion — when a competitor makes an acquisition, management feels pressure to respond strategically, perpetuating the wave.

M&A Avenues: Grow vs Optimise/Downsize

Q: What are M&A avenues: Grow vs Optimise/Downsize?

Objective	Action	Illustrative Deal Types
Grow	Add (+) — Pay less than value of business to the acquiror	Merger, Acquisition, Joint Venture
Optimise / Downsize	Restructure (-) — Sell for more than value of business to the vendor	Divestment/Sale, Carve-out, Private Placement, Spin-off, Split-off

Three Economic Perspectives on Mergers

Q: What are the 3 economic perspectives on mergers (horizontal, vertical, conglomerate)?

- **Horizontal:** Combination of two firms in the same industry and same stage of production. Primary goal: scale economies, market share, pricing power, elimination of duplicate functions. Example: two airlines merging.
- **Vertical:** Combination of a firm with its supplier (backward integration) or customer (forward integration). Goal: control of supply chain, reduced input costs, improved coordination.
- **Conglomerate:** Combination of firms in unrelated industries. Goal: diversification of cash flows, capital allocation across businesses. Empirically weak — tends to trade at a conglomerate discount of 10-20% relative to sum of parts.

Merger of Equals (MoE)

Q: What are the features of a Merger of Equals?

- Both parties are publicly traded
- Both parties have broadly similar market capitalisations
- Smaller party shareholders own no less than ~35-40% of the combined entity
- Consideration is typically entirely in stock (or very limited cash component)
- Nil to low premium paid — value creation comes from synergy sharing, not a control premium
- Less common than standard acquisitions but receives disproportionate attention

► *The Novozymes/Chr. Hansen deal was structured and communicated as a merger of equals — but the 49% premium to the free float and 38% blended premium suggest it functioned more as a disguised acquisition.*

Main M&A Deal Features

Q: What are the main M&A deal features?

Feature	Description
Consideration type	Cash, stock, mix, earnout, CVR
Premium	Price offered above target's unaffected market price
Friendly vs Hostile	Board-supported vs. opposed by board/management
Public vs Private	Listed target vs privately held company
Deal structure	Share deal, asset deal, or merger
Regulatory approvals	Antitrust/competition clearance required across jurisdictions
Financing	Equity, debt, bridge loans — mix depends on 6 drivers
Closing conditions	MAC clause, regulatory approvals, shareholder votes

M&A Participants

Q: Describe the various M&A participants and their roles.

Participant	Role in M&A
Board of Directors	Oversees strategy; approves/rejects deals; owes fiduciary duty to shareholders. In the US must apply Business Judgment Rule and Revlon duties where relevant.
Management	Executes deal; builds integration plan; subject to potential conflicts (empire-building, retention bonuses).
Financial Advisors / Investment Banks	Provide valuation, fairness opinions, negotiation support, and financing; earn fees on deal completion (binary).
Financing Banks	Provide bridge loans, acquisition debt, staple financing; separate from M&A advisory in large deals.
Legal / Tax / Antitrust Advisors	Draft/negotiate SPA, merger agreement; manage regulatory filings; assess tax implications.
Auditors	Validate financial statements; assist in financial due diligence.
Management Consultants	Commercial due diligence; synergy estimation; integration planning.
Proxy Solicitors	Engage shareholders ahead of votes; particularly relevant in contested situations.
Shareholders	Ultimate principals; vote on deal at EGM (merger) or tender shares (tender offer). Control vs minorities vs free float have different interests.
Sovereign Wealth Funds	Large passive or strategic LPs; potential white knights; long-term capital with non-commercial objectives at times.
Family Offices / HoldCos	Patient capital; strategic objectives may diverge from maximising short-term returns.
Hedge Funds / Activist Investors	Build minority stakes to push for M&A, governance changes, or capital returns; escalate from private to public campaigns.
Regulators	Antitrust/competition authorities (EC, CMA, DOJ/FTC); impose remedies or block transactions.
Debt Holders / Rating Agencies	Assess impact on credit quality; may restrict deal through covenants; rating agencies determine cost of new debt.
Employees, Clients, Suppliers	Affected stakeholders; integration risk; potential change-of-control clauses in contracts.
Press, Media, Analysts	Influence public perception; sell-side analyst reports shape institutional shareholder views.

CLASS 2 — M&A Process and Corporate Valuation

Deal Making Phase

Q: Describe the Deal Making phase and its 5 iterative workstreams.

The Deal Making phase (Phase C) encompasses Steps 5-8 and involves five iterative, simultaneous workstreams:

- **Deal Structuring:** Define consideration type (cash/stock/mixed), deal vehicle (merger/acquisition/asset deal), governance arrangements, and post-deal ownership.
- **Due Diligence:** Commercial, financial, and legal investigation to validate assumptions and identify risks.
- **Refine Valuation:** Continuously update valuation models as new information emerges from due diligence.
- **Financing Plan:** Structure and arrange acquisition financing — equity, debt, bridge — aligned with leverage tolerance, ratings, and timing.
- **Integration Plan:** Develop Day 1 and 100-day integration roadmap while deal is being negotiated.

► *These workstreams are iterative and interdependent — a due diligence finding changes valuation, which affects financing, which affects deal structure.*

Why Deal Synergies Matter

Q: Why do deal synergies matter?

Synergies are the economic rationale for M&A. A deal only has a positive economic basis if the two firms are worth more together than apart:

$$\text{NPV of Gain} = \text{PV}(\text{AB}) - \text{PV}(\text{A}) - \text{PV}(\text{B}) > 0$$

For the acquiror, value is created if the synergy gain exceeds the cost (premium paid):

$$\text{NPV to Acquiror} = \text{Synergy Gain} - \text{Premium Paid}$$

A buyer should never pay more than the NPV of synergies. If they do, value is transferred from the acquiror's shareholders to the target's shareholders — exactly what happened in Novozymes/Chr. Hansen where synergy NPV of ~EUR 1.5BN was dwarfed by the EUR 3BN free float premium.

Who Gets What: Value Appropriation

Q: What factors determine who gets what (value appropriation) between buyer and seller?

- Size of synergies and credibility — larger/more credible synergies allow higher premium while still being value-accretive to acquiror
- Number of competing bidders — auction drives up premium, benefiting seller
- Bilateral vs. auction process — bilateral gives buyer more pricing leverage
- Exchange ratio in all-stock deals — at-market ratio shares synergies equally; premium above market transfers value to seller
- Urgency and strategic necessity — the more the acquiror needs the deal, the more the seller captures
- Information asymmetry — seller knows business better; buyer compensates with due diligence and reps/warranties

Sources of Synergies

Q: What are the two main sources of synergies? Why are financing synergies less credible?

- **Cost Savings (Operating Synergies):** Shared IT/infrastructure, supply chain/production efficiencies, reduced headcount/board duplication, central functions, R&D consolidation, lower purchasing costs. Quantifiable and verifiable.
- **Revenue Benefits (Operating Synergies):** Complementary products, customers, territories; cross-selling; new growth opportunities; network externalities; new capabilities. More speculative and execution-dependent.
- **Financing Synergies:** Lower cost of borrowing post-combination due to greater scale and diversified cash flows. Less credible because: both companies can already access capital markets independently; any benefit may be immaterial or arises from increased leverage (which is a financial risk transfer, not true value creation); regulators and rating agencies see through purely financial restructuring.

► *Investors disproportionately focus on cost synergies and discount revenue synergies heavily — as demonstrated by the Novozymes market reaction to the EUR 200M revenue synergy announcement.*

Synergy Estimation: Top-Down vs Bottom-Up

Q: What are the top-down and bottom-up approaches to estimating deal synergies?

- **Top-Down (Market-Based):** Analyse synergies announced in comparable M&A transactions. Review ratios of previously announced synergies as a percentage of target revenues or EBITDA. Apply those ratios to the current deal to derive a range. Advantage: fast, market-calibrated. Disadvantage: may not reflect deal-specific dynamics.
- **Bottom-Up (Intrinsic):** Detailed analysis of acquiror and target standalone cost structures and revenue bases. Estimate potential savings/benefits for each revenue and cost line. Requires experienced line managers. Tailored to the specific combination. Factors in integration costs. Advantage: granular and deal-specific. Disadvantage: time-intensive; subject to optimism bias.

Seven Common Misconceptions About Valuation

Q: What are the 7 common misconceptions about valuation?

- **1. Valuations are objective:** All valuations embed assumptions. Two analysts with the same data can reach materially different conclusions depending on WACC, growth rates, and comparable selection.
- **2. Valuations are precise and accurate:** A DCF with many decimal places is still driven by highly uncertain long-term assumptions. Precision $\neq$ accuracy.
- **3. Valuations are timeless:** Market conditions, interest rates, and sector sentiment change continuously. A valuation is a snapshot in time.
- **4. More complex models lead to better valuations:** Complexity can obscure rather than clarify. The model follows the insight; it never replaces it.
- **5. Process of valuation is not important — only the outcome matters:** The discipline of the process (identifying assumptions, testing sensitivities) is where real understanding is built.
- **6. Markets are 100% efficient / 100% inefficient:** Neither extreme is correct. Markets are partially efficient most of the time but make systematic mistakes, especially in volatile periods.
- **7. Price = Value / Value = Price:** Price is what is paid in the market at a given moment. Value is the intrinsic worth based on fundamentals. They can diverge significantly in the short term.

Two Main Valuation Approaches

Q: Describe the two main valuation approaches: Relative (Pricing) vs Intrinsic (DCF).

Approach	Definition	When It Works Best
Relative / Pricing	Estimate value by observing what the market pays for comparable assets, scaled to a common metric (multiple). 'What are others paying for similar businesses?'	Large number of listed comparables; shorter investment horizon; market pricing trusted; useful for quick cross-checks.
Intrinsic / DCF	Present value of expected future free cash flows, discounted at a risk-adjusted rate. 'What cash flows will this business generate, and at what risk?'	Long investment horizon; analyst can provide catalyst; assets generate cash flows; investor not swayed by market noise.

Market Observed Valuation Benchmarks

Q: What are market observed valuation benchmarks?

- Listed firm prices (trading comparables) — minority market prices of publicly traded peers
- Sell-side analyst target prices — forward-looking intrinsic values from equity research
- LBO 'valuation' — the maximum price a PE firm will pay to achieve a target IRR given a specific leverage structure

LBO and Buyout Value Drivers

Q: What is an LBO? What are the three main buyout value drivers? Which is least predictable?

An LBO (Leveraged Buyout) is a buyout transaction where a target company is acquired using debt as the principal source of financing. PE firms borrow up to 70-80% of the purchase price from various lenders and fund the balance with equity from their funds. An MBO (Management Buyout) is an LBO where the target's own management team participates alongside the buyout firm. An MBI (Management Buy-In) is an LBO where an external management team acquires the business.

The three main buyout value drivers are:

- **1. EBITDA Impact:** Improve business operations — grow revenues and improve margins. Makes the business more valuable at any given multiple.
- **2. Multiple Effect:** Buy cheap (at a low EV/EBITDA multiple) and sell expensive. Entry vs exit multiple expansion is the most powerful driver but also the least predictable — entirely dependent on market conditions at exit.
- **3. Leverage Effect (Net Debt Contribution):** Use target's own cash flows to repay acquisition debt during the holding period. At exit, extract more equity from a lower net debt base.

► *The multiple effect (market re-rating) is the least predictable because it depends on market sentiment and macro environment at exit, which is entirely outside the PE firm's control.*

Two Approaches to Buyout Valuation

Q: Describe two approaches for buyout valuation: Fixed IRR vs Fixed Debt.

- **Fixed IRR Target:** Set a target return (e.g. 25% IRR) and a fixed leverage level. The resulting equity cheque defines the maximum bid price. If leverage changes, the bid price adjusts proportionally. Shows how debt level affects achievable price at the same IRR.
- **Fixed Debt Target:** Fix the debt quantum. Change the required IRR. Shows the trade-off between ambition (IRR target) and what you can bid. Lower IRR tolerance → higher bid. Higher IRR requirement → lower bid.

Illustrative comparison (from Session 2):

Approach	IRR Target	Debt	Equity	Bid
Fixed IRR	25%	\$440M	\$537M	\$977M
Fixed IRR	25%	\$550M	\$492M	\$1,042M
Fixed IRR	25%	\$660M	\$446M	\$1,106M
Fixed Debt	30%	\$550M	\$420M	\$970M
Fixed Debt	25%	\$550M	\$492M	\$1,042M
Fixed Debt	20%	\$550M	\$579M	\$1,129M

Relative Valuation: Pros, Cons, Steps

Q: What are relative valuation methods, their pros/cons, when do they work best, and what are the major steps?

Advantages	Disadvantages
Fast and simple to apply	Too simplistic — can miss key drivers
Fewer explicit assumptions required	Can miss important value-relevant variables
Easy to communicate to boards	Lack of transparency; prone to anchoring bias
Reflective of current market environment	If whole sector is overvalued, comps give overvalued answer
Price of 'today' — what a buyer can actually pay	Market can be collectively wrong

Works best when: many comparable assets exist; assets are priced in liquid markets; common variables allow standardisation; investor has shorter time horizon; investor judged against relative benchmark.

Major steps to applying relative valuation:

- **Step 1:** Define comparable companies universe — same cash flow profile, growth potential, and risk. Use sector, geography, and business model as filters.
- **Step 2:** Select appropriate multiple — EV/EBITDA, EV/EBIT, EV/Sales, P/E. Choose based on what drives value in the sector.
- **Step 3 (+1):** Assess whether to apply a premium (M&A context — control premium) or discount (IPO context — liquidity discount, minority discount).

Comparable Firms and Multiples

Q: What is a comparable firm? What is a P/E multiple? What is a companion variable?

A comparable firm shares similar cash flows, growth potential, and risk profile with the target being valued. Comparables are typically sourced within the target's business sector, geography, and size range. The practical challenge is how narrowly to define comparability — a shorter list is more precise but may miss the range; a longer list requires more statistical controls.

P/E Multiple = Share Price / EPS = Market Cap / Net Income. Companion variable for P/E = earnings growth rate (also expressed as PEG ratio = P/E / growth rate, to adjust for different growth profiles).

A companion variable is the single variable that most determines a multiple, extractable from a DCF model. The shortcut: divide net income by the denominator for equity multiples (e.g. net income / book value = ROE for Price/Book); divide after-tax operating income by the denominator for EV multiples (e.g. after-tax operating income / sales = after-tax operating margin for EV/Sales).

Enterprise Value Multiples

Q: List three enterprise value multiples.

- **EV/EBITDA:** Most commonly used. EBITDA is a proxy for operating cash flow before capex. Companion variable = after-tax operating income / EBITDA (reinvestment rate). Useful across capital structures since it is pre-interest.
- **EV/EBIT:** Useful when depreciation policies vary materially across comparables. Also exists as EV/EBITA (adds back amortisation of intangibles) and EV/EBITDAR (adds back rental costs — relevant for airlines, retail).
- **EV/Sales:** Useful for high-growth, unprofitable businesses where EBITDA is not meaningful. Companion variable = after-tax operating margin.

Precedent Transactions vs Trading Comps

Q: What are precedent transactions? How do they differ from trading comps?

Dimension	Precedent Transactions	Trading Comparables
What they measure	Control prices paid by acquirors for comparable businesses	Minority market prices of publicly traded peers
Premium	Include negotiated control premium (typically 20-40%)	Reflect minority/liquidity pricing without control premium
Timeliness	Historical — may predate current market conditions	Current market prices — reflect today's sentiment
Use case	Primary benchmark in M&A valuation — captures deal economics	Secondary benchmark / cross-check
Key risk	Deal-specific factors may make true comparability difficult	Market can be collectively overvalued or undervalued

DCF

Q: What is a DCF? What is its underlying philosophy? Pros, cons, and when does it work best?

A DCF (Discounted Cash Flow) values a business as the present value of its expected future free cash flows, discounted at a risk-adjusted rate:

$$BV = FCF1/(1+r) + FCF2/(1+r)^2 + \dots + FCFN/(1+r)^N + TV/(1+r)^N$$

Underlying philosophy: every asset has an intrinsic value based on its cash flows, growth, and risk. Markets make mistakes in pricing assets over time but correct themselves as new information becomes available. The DCF attempts to see through market noise to true fundamental value.

Advantages	Disadvantages
Based on fundamentals; not driven by market sentiment	Requires many explicit, difficult-to-estimate inputs
Forces confrontation of detailed assumptions	Highly sensitive to WACC and terminal growth rate
Less affected by market cycles	Subject to manipulation by the analyst
Useful if holding asset long-term to collect cash flows	No guarantee of over/under-valuation emerging
Can identify assets mispriced by the market	Problematic for sell-side analysts who need clear Buy/Sell signals

Works best for: firms that generate cash flows (not gold, currencies, collectibles); investors with long time horizons; investors who can provide a catalyst to realise value; investors not easily swayed by short-term market movements.

CLASS 3 — Due Diligence and Deal Execution

Due Diligence

Q: What is due diligence? In which 3 ways does it de-risk a deal? When is it done?

Due diligence is a process of enquiry and investigation made by a prospective buyer to confirm that it is buying what it thinks it is buying. It de-risks a deal in three ways:

- **1. Valuation:** Feeds information into the valuation model — e.g. confirming cost synergies, integration costs. Identifies items impacting sale price and provides data for negotiation.
- **2. Liabilities:** Uncovers risks and potential liabilities before closing. Identifies points requiring legal protection. Spots deal-breakers.
- **3. Integration:** Builds the foundation of the integration plan. Soft aspects (culture, management, retention) often as important as hard (systems, financials).

Timing: Outside-in due diligence happens before first contact (desk research, public information). Inside-out due diligence happens during the deal making phase when the target provides access to confidential information via VDR and management presentations.

Q: In what circumstances might it be acceptable not to do due diligence?

- Target is a public company and approach is unwelcome (no access granted)
 - Buyer knows the target or market extremely well
 - Target is small relative to buyer (materiality threshold)
 - Buyer relies exclusively on reps and warranties (though not recommended)
- *Not doing due diligence is never recommended — it is only occasionally acceptable, not ideal.*

Three DD Workstreams

Q: What are the three main due diligence workstreams?

Workstream	Focus	Primary Objectives	Supported By
Commercial DD	Market dynamics; competitive positioning; strategy; commercial prospects	Validate sustainability of future profits; confirm synergies; input for integration planning	Management consultants
Financial DD	Adequacy of accounting; validity of historical financials; quality of earnings	Confirm reported underlying profits; identify adjustments; inputs for valuation and warranties	Audit firm
Legal DD	Contractual agreements; litigation; problem-spotting	Validate existing contracts; identify liabilities; inputs for SPA/warranties	Law firm

Public M&A Acquisition Structures

Q: What are the two main forms of acquisition structures in US public M&A?

- **One-Step Merger:** Target shareholders vote at an EGM to approve the merger. Target entity is legally merged into acquiror. More common for friendly deals. Requires majority or supermajority shareholder vote. Takes longer but avoids the second step in tender offers.
- **Two-Step Merger / Tender Offer:** Acquiror makes a direct offer to target shareholders to tender (sell) their shares at a specified price. If successful (usually 50%+ acquired), acquiror then completes a back-end merger to acquire remaining shares at the same price. Faster than a one-step merger; can be used in both friendly and hostile contexts.

Special Committee

Q: What is a special committee? When is it relevant?

A special committee is a sub-group of independent (non-executive) directors formed to evaluate a transaction where the full board has a conflict of interest. Relevant when: management is on the buy side of an MBO; controlling shareholder is negotiating with the company (e.g. a take-private); a board member has a financial interest in the deal. In the US, a special committee can provide 'entire fairness' protection when approving a related-party transaction. The Novozymes/Chr. Hansen deal arguably warranted a special committee given Novo Holdings' dual ownership position — the absence of one was part of the governance concern.

Earnout and CVR

Q: What is an earnout? What is a contingent value right (CVR)?

- **Earnout:** A provision in the SPA where part of the purchase price is deferred and paid only if the target achieves specified financial milestones post-closing (e.g. revenue or EBITDA targets). Used to bridge valuation gaps between buyer and seller. Benefit: aligns seller incentives with post-close performance; protects buyer from overpaying. Risk: disputes over milestone definitions; management behaviour distortion.
- **CVR (Contingent Value Right):** Rights granted by the buyer to seller shareholders, typically event-driven, compensating sellers for positive future developments not yet reflected in the purchase price. Can be exchange-traded (separate securities) or non-transferable. In the Campari/Grand Marnier deal: the Villa Les Cèdres CVR guaranteed sellers the floor price of EUR 80M plus any excess above the floor from disposal proceeds. Unlike an earnout, a CVR is typically tied to an external event (asset disposal, clinical trial result) rather than operating performance.

Public M&A Legal Concepts

Q: Define: Business Judgment Rule, Revlon Duties, Frustrating Actions, Go-Shop, Fiduciary Out, Break Fee, Matching Right, Reverse Break Fee, MAC/MAE, Drop Dead Date, Disclosure Requirements.

Term	Definition
Business Judgment Rule	US corporate law doctrine protecting directors from personal liability for business decisions made in good faith, on an informed basis, with no conflict of interest. Shields boards from second-guessing if they followed proper process.
Revlon Duties	US doctrine triggered when a company is being broken up or sold to a third party. Board's duty shifts from long-term stewardship to maximising short-term sale price for shareholders. Once triggered, board must run a fair sale process.
Frustrating Actions	Actions by a target board that would frustrate or prevent a potential bid — e.g. issuing new shares, selling crown jewels. Under the UK Takeover Code, frustrating actions generally require shareholder approval once an approach is made or imminent.
Go-Shop Provision	Allows the target to actively solicit competing bids for a specified period after signing a merger agreement. Provides a market check. More common in PE deals.
Fiduciary Out	Clause permitting a target board to terminate a signed merger agreement if it receives a superior unsolicited proposal, without breaching its fiduciary duties to shareholders. Triggered after receiving a higher bid.
Break-Up / Termination Fee	Fee paid by the target to the acquiror if the deal is terminated by the target (typically due to accepting a superior offer). Compensates acquiror for deal costs. Typically 1-3% of deal value. In LVMH/Tiffany: \$575M.
Matching Right	Gives the acquiror the right to match any superior proposal received by the target during a go-shop or fiduciary out process, before the target can terminate in favour of the higher bid.
Reverse Break-Up Fee	Fee paid by the acquiror to the target if the acquiror fails to close (e.g. financing failure, regulatory block). Compensates target for walking away from other opportunities. Protects seller in situations where buyer has Hell or High Water obligation.

Term	Definition
MAC / MAE Clause	Material Adverse Change / Material Adverse Effect. Allows either party to exit before closing if a material adverse event occurs affecting target's business, financial condition, or operations. Excludes broad systemic events (acts of God, geopolitical conditions) but may apply if the event has a disproportionate impact on the target vs peers. LVMH attempted (and failed) to use this against Tiffany citing COVID.
Drop Dead Date	Termination date in the merger agreement beyond which either party can walk away if closing conditions have not been met. Provides certainty that the deal cannot run indefinitely. LVMH/Tiffany: 24 August 2020, unilaterally extendable 3 months to 20 November 2020.
Disclosure Requirements	In the UK (Takeover Code), once a company is the subject of an approach, it must disclose within specified timeframes ('put up or shut up' rule — 28 days to announce a firm intention or withdraw). In the US, SEC disclosure requirements govern when material information must be publicly disclosed.

Private M&A Transaction Structures

Q: What are the 3 basic transaction structures for private M&A? Which is most common?

Structure	Description	Advantages	Disadvantages
1a. Share/Stock Deal (most common)	Buyer acquires all shares of target entity. Entity, including all assets and liabilities, transfers to buyer.	Simple; no asset-by-asset transfer; preserves contracts and licences.	Buyer inherits all liabilities including unknown/contingent liabilities. Mitigated by reps & warranties.
1b. Merger	Target entity merges into acquiror or a new entity by operation of law. Shareholders of target receive consideration.	Clean legal structure; no asset transfer; single step.	Requires shareholder vote; more complex regulatory process.
2. Asset Deal	Buyer acquires specified assets (and selected liabilities) of the target, not the entity itself.	Buyer cherry-picks assets; avoids unwanted liabilities; potential tax step-up.	Complex to transfer each asset/contract individually; may require third-party consents; may trigger change-of-control clauses.

SPA and Purchase Price Mechanics

Q: What is an SPA? Purchase price adjustments? Escrow? Reps and warranties? Non-compete? Non-solicit? Hell or High Water?

- **SPA (Sale and Purchase Agreement):** The principal legal document governing a private M&A transaction. Sets out: purchase price, representations and warranties, conditions to closing, covenants, indemnification, and termination rights.
- **Purchase Price Adjustments:** Mechanisms to adjust the agreed price from signing to closing for changes in working capital, net debt, and transaction expenses. Normalises for timing differences between signing and closing. Most common: net debt adjustment (debt-free/cash-free basis) and working capital adjustment (against a normalised target).
- **Escrow:** A portion of the purchase price held back in a third-party account for a specified period post-closing. Released to seller only if indemnification claims do not materialise. Protects buyer against warranty breaches discovered after closing.
- **Reps and Warranties:** Representations are statements of fact about the target made by the seller as of a specified date (e.g. 'there is no material litigation'). Warranties are contractual assurances that those statements are true. Breach allows buyer to claim damages. R&W insurance increasingly common to replace or supplement seller's indemnification exposure.
- **Non-Compete:** Covenant preventing the seller from competing with the acquired business for a specified period and geography post-closing. Protects goodwill acquired by the buyer. Must be proportionate to be legally enforceable.
- **Non-Solicit / No-Hire:** Prevents seller from soliciting or hiring employees of the acquired business post-closing. Protects human capital that buyer has valued and paid for.
- **Hell or High Water:** Provision requiring the acquiror to take all actions necessary to obtain regulatory approval, regardless of cost (e.g. divesting assets). Shifts all antitrust risk onto the buyer. Used in deals with significant regulatory risk. In LVMH/Tiffany: LVMH assumed all antitrust risk.

CLASS 4 — M&A Financing

Drivers of M&A Financing Mix

Q: What are the 6 possible drivers of M&A financing mix?

- **1. Approach to leverage and leverage headroom:** Tolerance for debt varies by company, management, and shareholder. Key metric: Net Debt / EBITDA. Some boards have hard leverage caps (e.g. maintain investment grade). Post-deal leverage determines which financing instruments are feasible.
- **2. Credit rating impact:** Investment grade boundary (BBB-/Baa3) is critical. Crossing it dramatically widens spreads, shrinks investor universe, and may trigger covenant breaches. Boards typically solve for maintaining their rating when structuring M&A financing.
- **3. Funds certainty (regulatory and tactical):** Under the UK Takeover Code, a cash offer must be fully financed before announcement ('certain funds'). Tactically, in a competitive auction, demonstrated financing certainty is a differentiator. Drives use of bridge loans.
- **4. Confidentiality:** Approaching multiple banks for financing increases leak risk. Each additional bank multiplies the chance of a press leak, which could trigger regulatory disclosure obligations and allow competitors to react. Drives preference for bilateral bridge from the M&A advisor.
- **5. Timing:** Different instruments have materially different execution timelines. ABO: overnight. Bonds: weeks. Rights issue: ~3 months + shareholder subscription period. Bridge loans can be arranged in days and replaced with permanent financing over months.
- **6. Availability and pricing of alternative sources:** Market conditions determine what is accessible. In 2022, the HY bond market effectively closed, preventing PE LBOs from issuing bonds. In 2020, investment grade companies accessed debt at record low spreads. Availability and pricing determine the optimal mix at a point in time.

Three Ways to Raise Equity

Q: What are the 3 main alternative ways to raise equity financing?

- **Follow-On / Primary ABO (Accelerated Bookbuild):** Non-pre-emptive placement of new shares to institutional investors, typically overnight or intraday. No pre-emption rights for existing shareholders. Size limited to ~5-10% of existing shares. Discount ~4% to last market price. Fastest equity route. Example: Campari raised EUR 650M via ABO in January 2024 for the Courvoisier deal.
- **Rights Issue:** Pre-emptive offering: existing shareholders offered new shares in proportion to their holdings at a discount to market (TERP less typical discount of 17-40%). All shareholders can maintain their proportional ownership. Takes ~3 months preparation plus ~2 week subscription period. Often requires shareholder vote. $TERP = (N_{existing} \times \text{current price} + N_{new} \times \text{subscription price}) / (N_{existing} + N_{new})$.
- **Convertible Bond:** Hybrid instrument with debt features (coupon) and an embedded equity conversion option at a premium (typically 30-35%) to market. Low/zero coupon compensated by conversion option value. Investor base is specialised. Size typically limited to EUR 1-2BN — used as a top-up, not a primary instrument. Example: Campari raised EUR 550M mandatory convertible alongside the ABO.

Debt Instruments

Q: What is a revolver? What are differences between bonds and loans? What is a bridge loan and its advantages/disadvantages? What are advantages/disadvantages of bank debt?

- **Revolver (Revolving Credit Facility / RCF):** A committed line of credit that can be drawn and repaid freely, similar to a corporate credit card. Floating rate. Typically 3-5 year maturity. Not designed for permanent acquisition financing — expensive if drawn (banks charge higher rates to incentivise rapid repayment). Used for working capital and liquidity backstop. Every company of scale has one.

Dimension	Bonds	Bank Loans (Term Loans)
Rate	Fixed rate (almost exclusively)	Floating rate (SOFR/EURIBOR + spread)
Maturity	Long-dated: 5-30+ years	Typically 3-7 years
Repayment	Bullet at maturity; prepayment expensive (call premium)	Amortising (TLA) or bullet with periodic payments (TLB)
Covenants	Looser covenants (incurrence-based); 'cov-lite' HY	Tighter financial maintenance covenants in TLA; TLB often cov-lite
Investor base	Bond market: asset managers, insurers, pension funds	Bank market: commercial banks, CLOs, credit funds
Confidentiality	Requires rating, public disclosure	Can be arranged privately and confidentially
Size	Large size possible; deep market	Can be very large but bank approval process intensive

Bridge loan: short-term loan (typically <364 days), single purpose, floating rate. Arranged at announcement to provide financing certainty before permanent debt/equity is placed. Key characteristic: almost never actually drawn — the commitment is what matters. Very expensive if drawn (step-up margins incentivise rapid take-out). Provided by M&A advisor bank or commercial bank.

Bridge Advantages	Bridge Disadvantages
Speed of execution — can be arranged in days	Short-term only — must be replaced quickly
Removes financing contingency from deal	Fees (upfront commitment fee + step-up margins if drawn)
Confidentiality — can be bilateral with one bank	Bank approval process can be detailed and onerous
Large size capacity	Does not provide permanent solution
Does not cannibalize existing RCF or liquidity	

Bank debt advantages: lower cost (secured, covenants protect lender); structuring flexibility; no pre-payment penalties. Bank debt disadvantages: security/collateral required; more restrictive covenants (financial, affirmative, restrictive).

CLASS 5 — Business Separations and Sell-Side Process

Bilateral Negotiation vs Auction

Q: What is a bilateral negotiation? Advantages and concerns?

- **Definition:** A sale process where the seller engages with a single buyer exclusively, negotiating price and terms without a competitive bidding context.

Advantages: speed and confidentiality (fewer parties involved); appropriate when one logical buyer has materially higher synergies than all others; preserves relationship when cultural fit matters to seller; allows nuanced conversation about sensitive information without competitive risk. Concerns: no competitive tension limits price; seller cannot validate that maximum value has been achieved; creates fiduciary exposure if a higher offer later emerges.

Appropriate when: one buyer has a unique strategic rationale (e.g. only buyer without antitrust issues); the business story is complex and requires extended private engagement; founder prioritises cultural fit or legacy over price maximisation. Morgan Stanley data: approximately 36% of auction processes end with only one bidder anyway.

Two-Stage Auction Process

Q: What are the main phases of a two-stage auction process?

Phase	Duration	Key Activities
1. Preparation	8-10 weeks	Assemble advisors; define perimeter; prepare all deliverables (teaser, IM, VDD reports); identify buyer universe; define equity story.
2. Pre-Marketing (optional)	Variable	Warm-up Tier 1 buyers before formal launch ('Taylor Swift') or test market viability ('raccoon'). Senior level discussions. Plant seed of upcoming auction.
3. Round 1 — Non-Binding	4-6 weeks	Send teaser; execute NDAs; provide Information Memorandum and process letter; collect non-binding indicative offers with price range.
4. Round 2 — Binding	6-8 weeks	VDR access; VDD reports provided; management presentations; site visits; process letter 2; final binding bids with SPA markup.
5. Final Negotiations	2-4 weeks	Confirmatory due diligence; SPA negotiation; contract race; select preferred bidder.
6. Signing to Closing	TBD	Regulatory approvals; shareholder vote if required; closing. Can be same day (PE-to-PE) or 12-18 months (regulatory complex).

► Total typical duration: 5-7 months from kick-off to signing. Morgan Stanley real funnel: 29 pre-market → 14 Round 1 → 5 bids → 3 Round 2 → 2 final bidders.

Seller Objectives and Optimising Sale

Q: What are likely seller objectives? Key components to optimising a sale outcome? Key objectives of the preparation phase?

Seller objectives: (1) Maximise deal value — but not just headline price (certainty of close, tax efficiency, and rollover equity also matter); (2) Maximise certainty of close — prefer a certain deal at slightly lower price over a higher price with execution risk; (3) Minimise business disruption — protect confidentiality and operational continuity during process.

Three components to optimising sale outcome: (1) Meticulous preparation — front-load all due diligence, anticipate all material questions, prepare deliverables in advance; (2) Thoughtful process design — select right buyer universe, bilateral vs auction, timing relative to market conditions; (3) Rigorous execution — maintain competitive tension through multiple rounds, manage information release discipline.

Key preparation phase objectives: unmask and define key value blocks; assemble all required deliverables before launch; front-load due diligence topics to avoid surprises mid-process; ensure consistency — only positive surprises during process.

Key Auction Deliverables

Q: What are the key deliverables in an auction?

Deliverable	Phase	Description
Teaser	Before Round 1	3-5 landscape pages. Company overview, investment highlights, summary financials. No forward projections. No company name (confidential). Provided to buyers before NDA.
NDA (Non-Disclosure Agreement)	Before Round 1	Legal commitment to confidentiality. 3-6 days to negotiate. Includes standstill for public targets. Enables provision of confidential information.
Information Memorandum (CIM)	Round 1	75-150 pages. Primary marketing document. Business plan (3-5 years), detailed financials, market analysis, management profiles. Written to present best view (not mislead — that would be criminal).
Process Letter (Round 1)	Round 1	Sets rules of engagement: what must be in a Round 1 bid (price, funding, required approvals, deal structure, what else is needed). Asks for 100% price, not a range.
Staple Financing	Round 1 (optional)	Committed financing arranged by seller's bank for bidders. Levels playing field, especially for PE. Creates financing trees (separate teams per bidder with information barriers).
VDR (Virtual Data Room)	Round 2	Digital data room with full documentation. Provides daily usage reports showing which documents bidders reviewed. Index prepared by M&A advisor.
VDD Reports (Vendor Due Diligence)	Round 2	Financial, commercial, and legal reports commissioned by seller and provided to buyers. 5-6 weeks to prepare. Critical for PE buyers (need for investment committee approval).
Management Presentation	Round 2	Full morning session. Management presents business, strategy, financials, and plan. Interactive Q&A. Key for assessing management and cultural fit.
Process Letter (Round 2)	Round 2	Sets requirements for final binding bid: must include SPA markup in executable form, financing confirmation, all approvals obtained.
Draft SPA	Round 2	Provided to bidders at start of Round 2. Final bid must include buyer's markup. Enables a 'contract race' to closing.

Strategic vs Financial Buyers

Q: What are the two main types of buyers and their main differences?

Dimension	Strategic Buyer	Financial Buyer (PE)
Primary motivation	Synergies (cost + revenue), geographic expansion, defensive consolidation	Financial return (IRR); operational improvement; leverage paydown
Ability to pay	Can pay control premium justified by synergies	Limited by leverage and target IRR — disciplined bidders
Financing	Balance sheet, equity issuance, bonds/loans	Primarily leverage (LBO structure); 70-80% debt
Due diligence	Lighter (more industry knowledge)	Heavier ('losing is not an option')
Management	Often replace with own team; management tenure risk	Typically retain and incentivise management (MBO/MBI)
Confidentiality risk	Competitor knows your strategy if deal fails	Lower risk — PE funds are not operational competitors
Exit horizon	Permanent ownership (strategic integration)	3-7 years — must have clear exit route

Process Design Parameters

Q: What are the 3 main parameters driving process design?

- **1. Buyers Universe:** How broad to cast the net? Strategic vs PE vs both? Geographic spread? Degree of bidder familiarity with the asset? Pre-marketing relevant to educate? Level playing field creation?
- **2. Timing:** Typical annual breaks (summer, Christmas). Broader market conditions — comparable company performance, recent M&A precedents. Amount of preparation needed — VDD reports, equity story definition, business plan milestones.
- **3. Transaction Certainty:** Is a broad auction a safety net? Are financial sponsors competitive (depends on financing availability)? Is recapitalisation a credible alternative? Is IPO/spin-off a dual-track option?

Pre-Marketing, Staple Financing, VDR, Dual Track

Q: What are the two possible objectives of pre-marketing? What is staple financing and its conflicts? What is a VDR? What is a dual track?

- **Pre-Marketing objectives:** (1) Build-up/warm-up phase — generate excitement, position the asset as highly sought, make Tier 1 buyers feel privileged ('Taylor Swift'); (2) Putting feelers out — test market viability, get directional price feedback, identify whether the business can be sold at all ('raccoon' check).
- **Staple Financing:** Financing package arranged by the seller's investment bank for all potential bidders simultaneously. Bidders can use it or find their own financing. Advantage: levels playing field; speeds up process; provides price/confidence signal; maximises auction competition. Conflict of interest: the seller's bank is providing financing to the buyer — it has an incentive to lend generously to facilitate a deal, which may not be in the lender's best long-term interest. Bank creates 'financing trees' (separate teams per bidder) to manage information barriers.
- **VDR (Virtual Data Room):** Secure digital platform containing all due diligence materials. Key feature: provides daily usage tracking reports showing which bidders looked at which documents — allows seller's advisor to gauge real interest and strategy.
- **Dual Track:** Running an M&A sale process and an IPO process simultaneously. Forces both processes to compete. IPO disciplines M&A bidders (demonstrates standalone value); M&A process disciplines IPO bankers (creates competitive urgency). The seller selects whichever outcome is superior at decision time. Costly and operationally demanding but can maximise value.

Business Separations

Q: What are the four main business separation types? Similarities and differences?

Type	Separation	Proceeds to Parent?	New Entity Listed?	New Shareholders?	Achieves
Sale	Full (100%)	Yes — full price + control premium	No	New owner 100%	Maximum cash proceeds; complete exit
Carve-out / Spin-IPO	Partial (~25-30% via IPO)	Yes — fraction (IPO proceeds)	Yes	Institutional investors via IPO	Retains majority (70%+); unlocks valuation; creates listed currency
Private Placement	Partial (minority stake)	Yes — fraction (private)	No	1-2 specific private investors	Capital without public disclosure; strategic partner
Spin-off / Demerger	Full	No cash to parent	Yes (SpinCo separately listed)	Existing parent shareholders receive SpinCo shares	Full strategic separation; unlocks conglomerate discount; no cash raised
Split-off	Full	No cash to parent (shares cancelled)	Yes	Specific shareholders exchange parent shares for SpinCo shares	Targeted separation; reduces parent share count

Spin-off advantages: unlock conglomerate discount (10-20%); separate investor bases; management focus; strategic clarity; pure-play premium (+25-45% per Morgan Stanley research). Spin-off challenges: flowback risk (index selling); reduced analyst coverage; takes 2+ years to execute (Novartis/Alcon: 2+ years; Novartis/Sandoz: 2+ years); one-off costs; business disruption.

Four driver buckets for business separations: (1) Company drivers — proactive (different financial profiles, capital optimisation, management focus); (2) Company drivers — defensive (liability ring-fencing, raise capital urgently); (3) Market drivers — proactive (unlock conglomerate discount, passive indexation premium); (4) ESG (separate carbon-intensive assets).

Shareholder activism increasingly a fifth driver.

Private Placement Minority Rights

Q: What minority rights types do private placement investors ask for?

Bucket	Rights
Corporate Governance Rights	Board seat appointment and removal rights; board meeting frequency; top management appointments; rights of information and access; approval/consent rights on major decisions; related party transactions; litigation settlement; executive remuneration; contract setting; investments/capex; leverage; dividend policy; capital structure changes; material changes to business plan; annual budget approval.
Exit Rights	Lock-up period; sale notice requirements; pre-emption rights (right of first refusal on new share issuances); tag-along rights (right to sell alongside majority if majority sells); drag-along rights (majority can force minority to sell in a qualifying transaction); put/call options; IPO rights; other transfer restrictions.

CLASS 6 — Private Equity and M&A

LPs and GPs

Q: What are Limited Partners (LPs) and General Partners (GPs) in private equity?

- **General Partners (GPs):** The PE firm itself. Responsible for fund management, deal sourcing, execution, and portfolio management. Bear unlimited liability. Earn management fees and carried interest. Partners who own carry are the GPs — analysts and associates are employees until promoted to partner level.
- **Limited Partners (LPs):** The capital providers. Liability limited to capital committed — cannot lose more than invested. Typical LPs: pension funds, sovereign wealth funds, asset managers, endowments, insurance companies, family offices. Increasingly retail investors are being offered access through semi-liquid vehicles.
 - *Fund structure: closed-end limited partnership with ~10-year finite lifespan. LPs commit capital to a 'blind pool' — they do not know specific investments in advance. Capital is drawn down as needed ('capital calls') rather than paid upfront.*

Four Fund Phases

Q: What are the four main phases of a PE fund's lifecycle?

- **1. Fundraising:** GPs set target fund size and hard cap. Series of fund closings. Year of first closing = fund 'vintage.' GPs spend significant partner time fundraising. Once ~75% invested, begin raising next fund.
- **2. Investment Period:** Typically 4-5 years from first closing, extendable 1-2 years with LP consent. GPs draw down committed capital as investments are made. No new portfolio company investments permitted after period ends. Dry powder = committed but uninvested capital.
- **3. Holding Period:** Typically 3-7 years. GPs add value through add-on acquisitions, operational improvement, repositioning, and capital structure optimisation. Median EMEA holding period reached 6.1 years in 2025 — highest ever.
- **4. Divestment / Exits:** Proceeds distributed to LPs after each exit. Exit routes: strategic sale, sponsor-to-sponsor, IPO, secondary. EMEA trend (2025): strategic exits 50%, sponsor-to-sponsor 33%, IPO 10%. GPs must exit to return capital to LPs and demonstrate performance for next fundraise.

Management Fees vs Carried Interest

Q: How do management fees differ from carried interest?

Dimension	Management Fee	Carried Interest ('Carry')
What it is	Annual fee paid by LPs to GPs for fund management	GPs' share of net fund profits
Typical rate	~1.3-1.5% (buyout); ~2% (VC)	~20% of net profits
When paid	Annually, regardless of fund performance	Only after LPs receive all capital back plus hurdle rate return
Purpose	Covers GPs' operating costs, salaries, infrastructure	Primary financial incentive to generate returns
Hurdle rate	N/A	~8% minimum return to LPs before carry kicks in
Catch-up	N/A	After hurdle, GPs receive catch-up allocation until at 20%
Long-term split	N/A	80% LP / 20% GP above hurdle after catch-up

► *Additional PE income: transaction fees and monitoring fees charged directly to portfolio companies. These go to GP partners, not the LP pool — a governance controversy.*

MBO vs MBI and Buyout Characteristics

Q: How do MBOs and MBIs differ? What are the three main characteristics of buyouts?

- **MBO (Management Buyout):** The target company's existing management team participates alongside a PE buyout firm to acquire the company through a leveraged structure. Management typically contribute meaningful personal equity and are granted an incentive pool.
- **MBI (Management Buy-In):** An external management team (brought in from outside) acquires the company alongside a PE firm via a leveraged structure. Used when incumbent management is not viewed as capable of delivering the value creation plan.

Three main characteristics of buyouts:

- **1. Majority Control:** PE firm acquires a controlling stake (>50%). Unlike VC (minority), buyouts give PE full governance control to drive operational change without needing other shareholders' consent.
- **2. Leverage:** Debt is the primary financing source (up to 70-80% of purchase price). Amplifies equity returns if the business performs. Imposes financial discipline. Cash flows used to repay debt, increasing equity value.
- **3. Management Alignment:** Management team incentivised through equity ownership (rollover or granted stake). Aligns management interests with PE financial objectives. Creates co-investment dynamic.

CLASS 7 — Public M&A, Hostile Takeovers and Activism

Activist Shareholders

Q: What is an activist shareholder? What are typical demands? What is the possible playbook?

An activist shareholder is a specialised hedge fund (or occasionally an institutional investor) that builds a minority stake in a publicly listed company to promote a specific agenda aimed at increasing shareholder value or improving ESG practices. Critically, activists do not seek control — they seek to influence board and management decisions.

Typical demands: 47% of campaigns involve an M&A-related agenda (sell the company, spin off a division, break up a conglomerate); 43% involve broader governance demands (board composition, CEO replacement, capital returns). Usually a combination of both.

Activist playbook (sequential escalation):

Private phase:

- Develop investment thesis with specific catalysts
- Research: speak to competitors, employees, customers, suppliers to confirm thesis
- Assess likelihood of support from other shareholders
- Accumulate equity through direct purchases and derivatives — avoid disclosure thresholds as long as possible
- Approach management privately with demands
- Request private board meeting to present thesis
- (May) request board seat privately

Public phase (if private engagement fails):

- Publicly release detailed analysis via letter, presentation, and dedicated website
- Publicly criticise board and management performance
- Discredit current board as entrenched and ignoring shareholders
- Nominate opposing slate of directors (primarily US tactic)
- Attempt to oust CEO or key management
- Engage in litigation
- Work with strategic partners to crystallise a sale or offer

Hostile Takeover

Q: What is a hostile takeover? What is the distinction between unsolicited and hostile?

Key distinction: Unsolicited = bidder makes an approach the target did not request, but target board has not yet rejected it. Hostile = target has said no and bidder continues anyway.

A hostile takeover is: (i) an acquisition of control of a publicly traded company; (ii) through an unsolicited approach and/or direct tender offer to shareholders; (iii) without initial consent of target board/management; (iv) contested by the board; (v) where control ultimately changes hands. Hostile tactics are legitimate (not illegal) — but discipline is required: overpaying leads to buyer's remorse and difficult returns. Frequency: ~7-10% of global M&A volumes.

Four Knights

Q: What is a 'black', 'yellow', 'white' and 'grey' knight?

- **Black Knight:** The entity driving the hostile approach — the unsolicited/unwelcome bidder.
- **Yellow Knight:** A black knight that eventually backs out of its hostile attempt and instead proposes a friendly combination or walks away entirely.
- **White Knight:** A third party that makes a friendly, welcome offer to the target in the presence of a black knight — giving the target board an alternative to the hostile approach.
- **Grey Knight:** A fourth party that makes an alternative bid higher than the white knight, creating a competitive dynamic between the white and grey knights.

Defence Mechanisms

Q: What are golden parachutes, crown jewel provisions, standstill provision, and defence manual?

- **Golden Parachutes:** Generous severance arrangements for senior executives triggered by a change of control — typically 2-3x annual compensation plus accelerated vesting of equity. A shark repellent pre-offer defence. Makes an acquisition more expensive by increasing the cost of replacing management. Also provides management with some protection against self-interested anti-takeover behaviour by aligning their financial interests with a successful deal.
- **Crown Jewel Provisions:** Allow the board to sell key assets ('crown jewels') to a third party if a takeover attempt is imminent, making the target less valuable and therefore less attractive to the hostile bidder. A scorched earth tactic. Generally restricted under the UK Takeover Code post-approach without shareholder consent.
- **Standstill Agreement:** A negotiated agreement whereby a potential bidder agrees not to increase its stake in the target above an agreed threshold for a specified period. Can be a pre-emptive measure or negotiated after an initial approach to pause hostilities while discussions are conducted. Note: under the UK Takeover Code, many post-offer defences (greenmail, restructurings) require shareholder approval — the regime strongly protects shareholder rights to decide on offers.
- **Defence Manual:** A company's pre-prepared playbook for responding to an unsolicited or hostile approach. Prepared in advance, not in crisis mode. Contains: who gets called first (lawyers, board, PR); what language to use in initial response; valuation framework to be applied; which advisors to retain; what tactical options are available; how to handle information requests; shareholder communication plan. The Britvic board's response to Carlsberg in Session 7 illustrated a live defence manual in action.

CLASS 8 — Mergers and Associated Concepts

Purchase Price vs Consideration Paid

Q: What are the differences between purchase price and consideration paid?

- **Purchase Price:** The agreed price for 100% of the target's equity — what appears in press releases and announcements. This is the equity value being paid. For EV, you add net debt and other balance sheet items.
 - **Consideration Paid:** The actual instruments used to pay the purchase price — cash, stock, convertibles, earnouts, CVRs, or a mix. The consideration is how the purchase price is funded and delivered to the seller's shareholders. Distinctions matter because: all-cash consideration is certain; all-stock consideration has a floating value linked to acquiror's share price; mixed deals have different risk allocation between buyer and seller.
- In the Novozymes/Chr. Hansen deal: the purchase price (implied by the exchange ratio) was approximately EUR 10.85BN, but the consideration was entirely in Novozymes B-shares (1.5326 per free float CHR share; 1.0227 for Novo Holdings).

Key Merger Deal Elements and Soft Issues

Q: What are the key deal elements to consider in a merger? Which are 'soft' issues?

Hard (financial/structural) elements: relative valuation; deal pricing; financial impacts (EPS accretion/dilution); estimated synergies; value creation analysis; pro-forma capital structure; rating impacts; ongoing shares listing location; flowback; transaction structure; surviving entity; domicile and HQ; controlling shareholders; commitments and expectations; ongoing representation.

Soft ('flag') issues — often emotionally charged and key to deal success:

- Enlarged group strategy and vision
- Marketing name, logo and branding, legal name
- Combined board composition
- Combined group management
- CEO appointment
- Management composition and decision-making structure
- Succession plans

► Soft issues are frequently deal-breakers or deal-makers in mergers of equals. They can also disguise power dynamics — in Novozymes/Chr. Hansen, Novozymes retained the CEO, CFO, entity continuity, and Chairman nomination, suggesting it was the dominant party despite 'merger of equals' framing.

Merger of Equals Challenges

Q: What challenges can a merger of equals face?

- Governance gridlock — equal representation on the board slows decisions; no clear majority
- CEO succession — typically one CEO is named 'Co-CEO' or a rotation is agreed, both of which create uncertainty
- Cultural clash — different corporate cultures, geographies, and management styles are difficult to integrate without clear authority
- Valuation disputes — agreeing an 'at market' exchange ratio that both sides accept as fair requires significant negotiation
- Synergy attribution — disagreements about who 'owns' the synergy value and who gets credit
- Flowback risk — in cross-border mergers, index funds may sell the target's shares post-merger if combined entity does not qualify for target's home index
- Management retention — uncertainty during protracted negotiations leads to talent attrition

Headline Premium

Q: What is the headline premium in a deal? How is it calculated?

The headline premium is the percentage excess of the offer price over the target's last unaffected publicly traded share price before announcement.

Cash deal: $\text{Premium} = (\text{Offer Price per Share} / \text{Target Last Traded Price}) - 1$

Example: Offer at \$30 cash for a target trading at \$20: $\text{Premium} = \$30/\$20 - 1 = 50\%$

Stock deal: First calculate the implied value of the stock consideration using the acquiror's unaffected share price. Then compare to target's unaffected price.

Example: Acquiror shares at \$40, offering 1.25x per target share → Implied value = $1.25 \times \$40 = \50 . Target last traded at \$45. $\text{Premium} = \$50/\$45 - 1 = 11.1\%$

Important: The premium can be negative (a discount). Example: offering \$18 for a \$20 target share = -10% premium (10% discount). This sometimes occurs in distressed situations or where the offeror believes the target is overvalued.

In the Novozymes/Chr. Hansen deal: exchange ratio $1.5326 \times \text{NZY unaffected price DKK } 431 = \text{DKK } 660.55$ implied per CHR share. CHR unaffected price DKK 444.20. $\text{Premium} = 660.55/444.20 - 1 = 48.7\% \approx 49\%$.

Exchange Ratio and At-Market Exchange Ratio

Q: What is the merger exchange ratio? What is the 'at market' exchange ratio and what does it achieve?

The merger exchange ratio specifies how many acquiror shares each target share is exchanged for in a stock-for-stock transaction. Example: exchange ratio 1.5326 means each Chr. Hansen share receives 1.5326 new Novozymes B-shares.

The at-market exchange ratio is the ratio that exactly replicates the relative market values of the two companies:

$\text{At-Market Exchange Ratio} = \text{Target Share Price} / \text{Acquiror Share Price}$

Mario/Luigi example: Luigi Corp shares at \$50, Mario Inc shares at \$60 → $\text{At-market ratio} = \$50/\$60 = 0.833x$

What the at-market ratio achieves:

- No premium is paid to the target shareholders
- Both sides receive shares proportional to their current market values
- Without synergies: no value creation or destruction for either side
- With synergies: both sides share the synergy value equally in percentage terms — this is the defining property of the at-market ratio

► Any exchange ratio above the at-market ratio implies a premium. The higher the premium, the more value is transferred from acquiror shareholders to target shareholders. If synergies do not justify the premium, acquiror shareholders lose.

EPS and EPS Accretion/Dilution

Q: What is EPS? How do you calculate EPS accretion/dilution for the buyer in an all-stock merger?

EPS (Earnings Per Share) = Net Income / Number of Shares Outstanding

In an all-stock merger, the acquiror (surviving entity) issues new shares as consideration. The enlarged share count relative to the combined net income determines whether EPS increases (accretion) or decreases (dilution) compared to standalone:

Combined EPS = (Acquiror Net Income + Target Net Income $\pm$ Synergies) / (Acquiror Shares + New Shares Issued)

EPS Accretion / Dilution = (Combined EPS / Standalone Acquiror EPS) - 1

Mario Inc. / Luigi Corp. worked example (from Session 9):

Scenario	Calculation	Result
Standalone Mario EPS	\$300 net income / 100 shares	\$3.00
At-market ratio (0.833x): new shares issued	60 Luigi shares $\times$ 0.833 = 50 new Mario shares	Total = 150 shares
Combined EPS (no synergies, at-market)	$(\$300 + \$200) / 150$	\$3.33
EPS accretion at-market	$(\$3.33 / \$3.00) - 1$	+11.1% ACCRETIVE
Premium scenario (1.0x ratio): new shares issued	60 Luigi shares $\times$ 1.0 = 60 new Mario shares	Total = 160 shares
Combined EPS (no synergies, with 20% premium)	$(\$300 + \$200) / 160$	\$3.125
EPS accretion with 20% premium	$(\$3.125 / \$3.00) - 1$	+4.2% ACCRETIVE (less)

► *EPS fallacy: EPS accretion does not equal value creation. A higher P/E acquiror (Mario at 20x) buying a lower P/E target (Luigi at 15x) will always be mechanically EPS accretive at the at-market ratio regardless of synergies — the P/E blending effect alone causes it. The correct measure of value creation is the shift in shareholder wealth, not the EPS impact.*

Value Creation Calculations for Both Sides

Q: From the Novozymes case: how do you calculate value creation for both sides of a merger?

Step 1 — Establish pre-deal equity values for each party:

Party	Pre-Deal Value (EUR BN)
Novozymes free float (74.4% of NZY EUR 16.3BN)	12.1
Novo Holdings (25.6% of NZY)	4.2
Chr. Hansen free float (78% of CHR EUR 7.9BN)	6.2
Chr. Hansen Novo Holdings stake (22%)	1.7
Total combined	24.2

Step 2 — Establish post-deal ownership (from exchange ratios and press release):

- Novozymes free float: 44% of combined
- Novo Holdings: 22% of combined
- Chr. Hansen free float: 34% of combined

Step 3 — Calculate value creation under two scenarios:

Scenario A: No synergies (combined = EUR 24.2BN):

Party	Before	After (44%/22%/34% of 24.2)	Change
NZY free float	EUR 12.1BN	EUR 10.7BN	-12%
Novo Holdings	EUR 5.9BN	EUR 5.3BN	-10%
CHR free float	EUR 6.2BN	EUR 8.2BN	+32%

Scenario B: Full synergies NPV EUR 2.0BN (combined = EUR 26.2BN):

Party	Before	After (44%/22%/34% of 26.2)	Change
NZY free float	EUR 12.1BN	EUR 11.5BN	-5%
Novo Holdings	EUR 5.9BN	EUR 5.7BN	-2%
CHR free float	EUR 6.2BN	EUR 8.9BN	+44%

Conclusion: Even with full synergy delivery (EUR 2BN NPV), Novozymes shareholders lose 5% and Novo Holdings loses 2%. Chr. Hansen free float gains 44%. The synergies are insufficient to overcome the value transfer embedded in the 49% premium. This is the core market reaction rationale.

Key formula: Value creation for each party = (Post-deal ownership % × Combined Value) - Pre-deal value. At the at-market exchange ratio, this equals the party's ownership % × synergy NPV for both sides.

— End of Study Guide Answers —